

2.3 Competitive market equilibrium

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Learning objectives

2.3 Competitive market equilibrium	Depth	Diagrams and calculations
Demand and supply curves forming a market equilibrium	AO4	Diagram: market equilibrium
Shifting the demand and supply curves to produce a new market equilibrium, with reference to excess demand and excess supply	AO2 AO4	Diagram: showing changes in equilibrium/role of price mechanism
Functions of the price mechanism • Resource allocation (Signalling, Incentive) • Rationing	AO2	

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Learning objectives

2.3 Competitive market equilibrium	Depth	Diagrams and calculations
Consumer and producer surplus	AO2 AO4	Diagram: showing consumer surplus and producer surplus (social / community surplus) – maximized at competitive market equilibrium **Calculation (HL) only: Consumer surplus and producer surplus from a diagram**
Social/ community surplus	AO2 AO4	
Allocative efficiency at the competitive market equilibrium: • Social/community surplus maximized at equilibrium • Marginal benefit (MB) equals marginal cost (MC)	AO2 AO4	

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Real world example

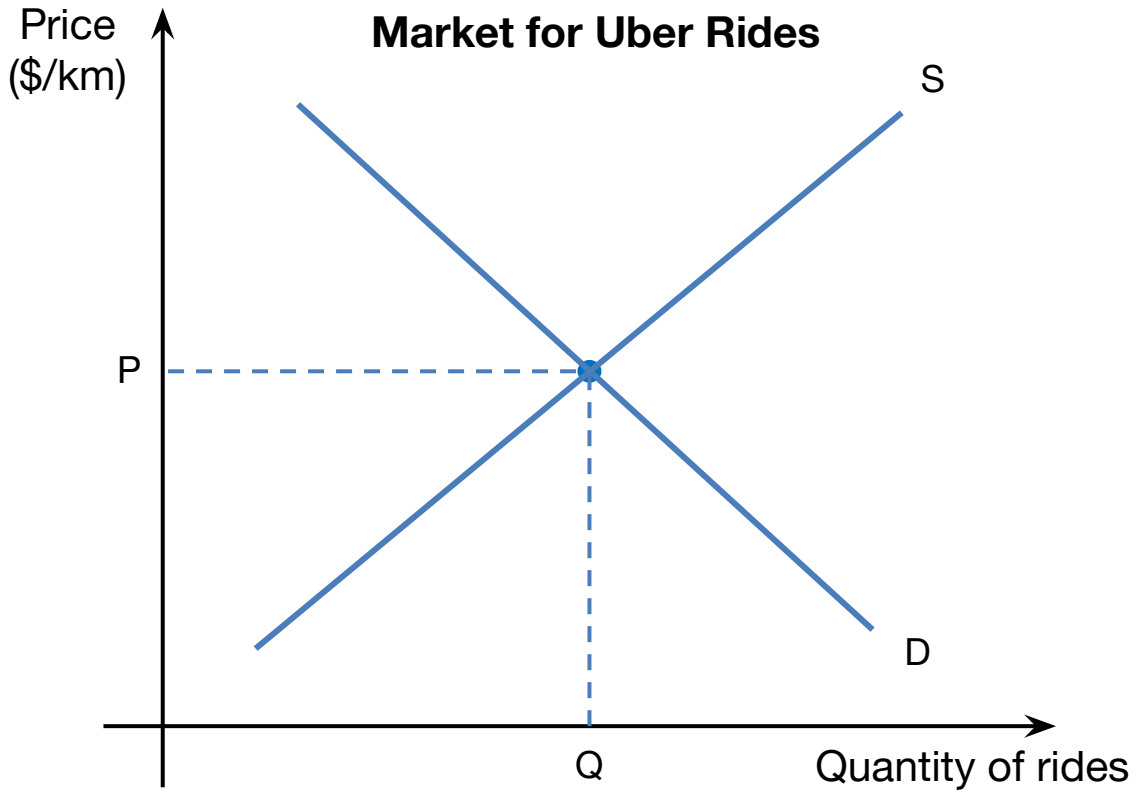
How might the Uber market be related to demand, supply, and market equilibrium?

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Market equilibrium introduction

Market equilibrium occurs at the price where quantity demanded equals to quantity supplied.

At this point, the market is cleared of any shortage or surplus.



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Market equilibrium – activity

The market demand and supply of Uber rides is displayed below.

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
20	4,000	8,000
15	6,000	6,000
10	8,000	4,000
5	10,000	2,000

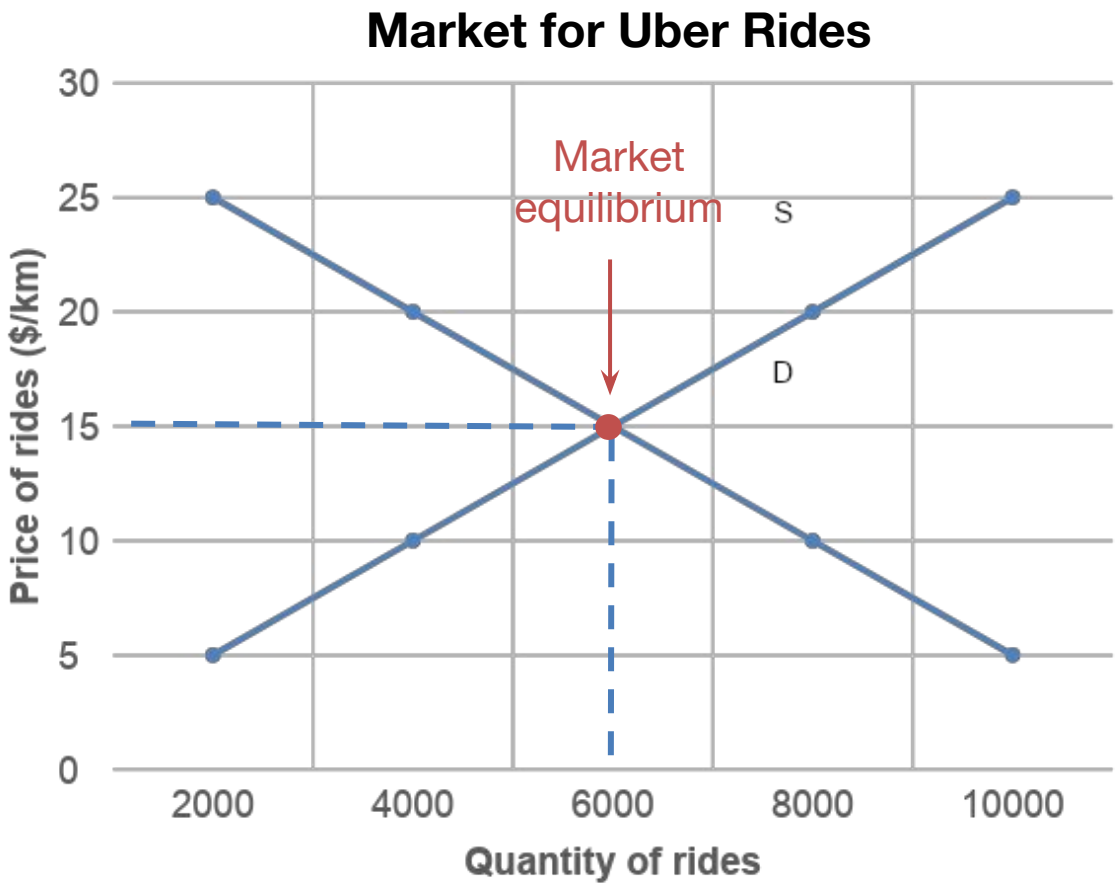
1. Plot the demand and supply curves.
2. Identify the equilibrium price & quantity.
3. What happens if the price is $10/km?
4. What happens if the price is $25/km?

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Market equilibrium – activity

1. Plot the demand and supply curves.

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
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Market equilibrium – activity

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
20	4,000	8,000
15	6,000	6,000
10	8,000	4,000
5	10,000	2,000

2. Identify the equilibrium price and quantity.

   Equilibrium price = $15

   Equilibrium quantity = 6,000 rides

3. What happens if the price is $10/km?

   Market shortage of 4,000 drivers.

4. What happens if the price is $25/km?

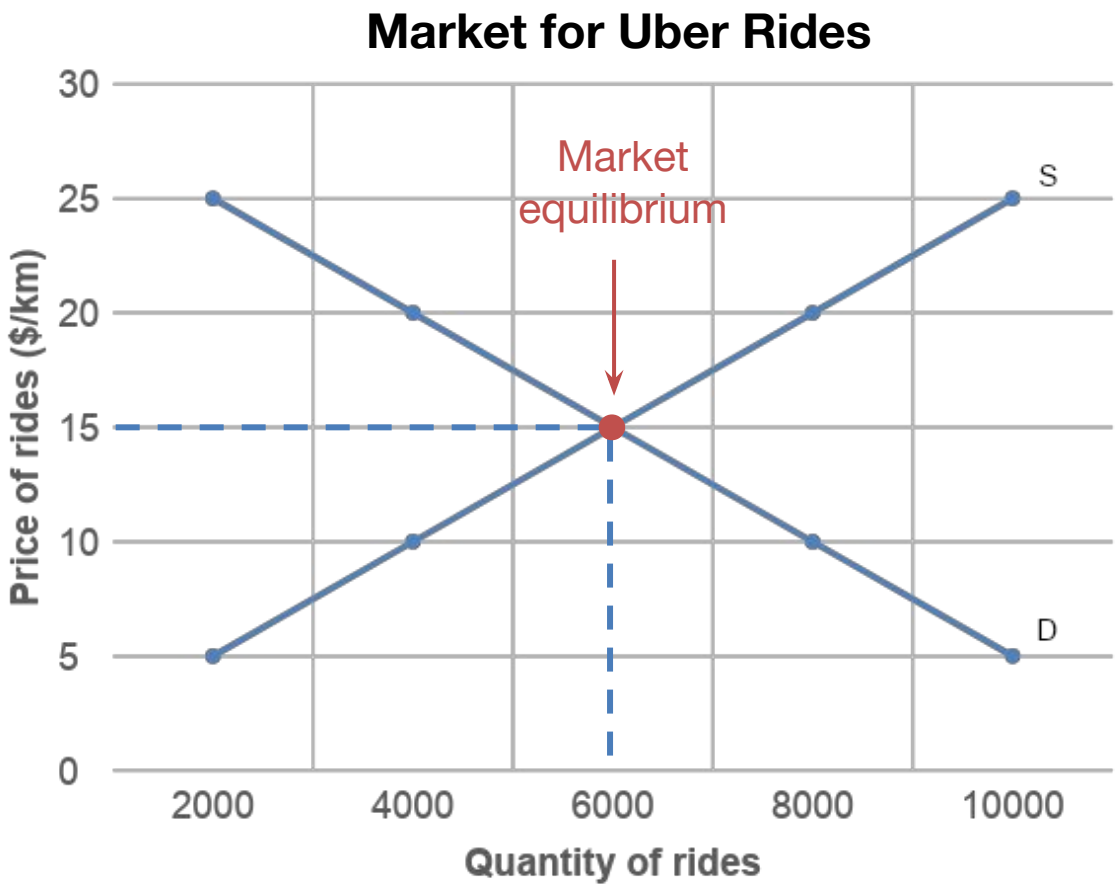
   Market surplus of 8,000 drivers.

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Market equilibrium

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
20	4,000	8,000
15	6,000	6,000
10	8,000	4,000
5	10,000	2,000

At the **market equilibrium**, there are no surpluses or shortages. The market "clears" where every rider is matched with a driver and vice versa.

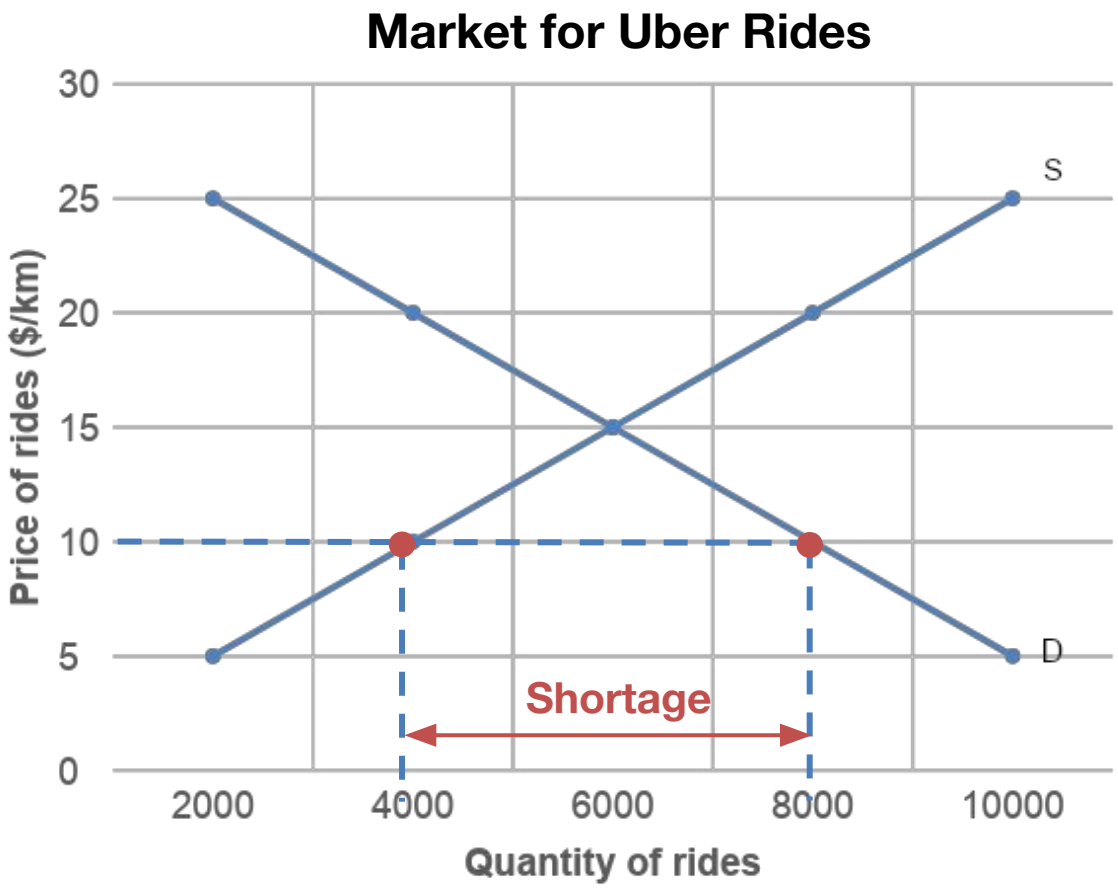


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Shortage – excess demand

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
20	4,000	8,000
15	6,000	6,000
10	8,000	4,000
5	10,000	2,000

If the price is lower than the equilibrium price, the quantity demanded is greater than quantity supplied, resulting in **excess demand (shortage)**.

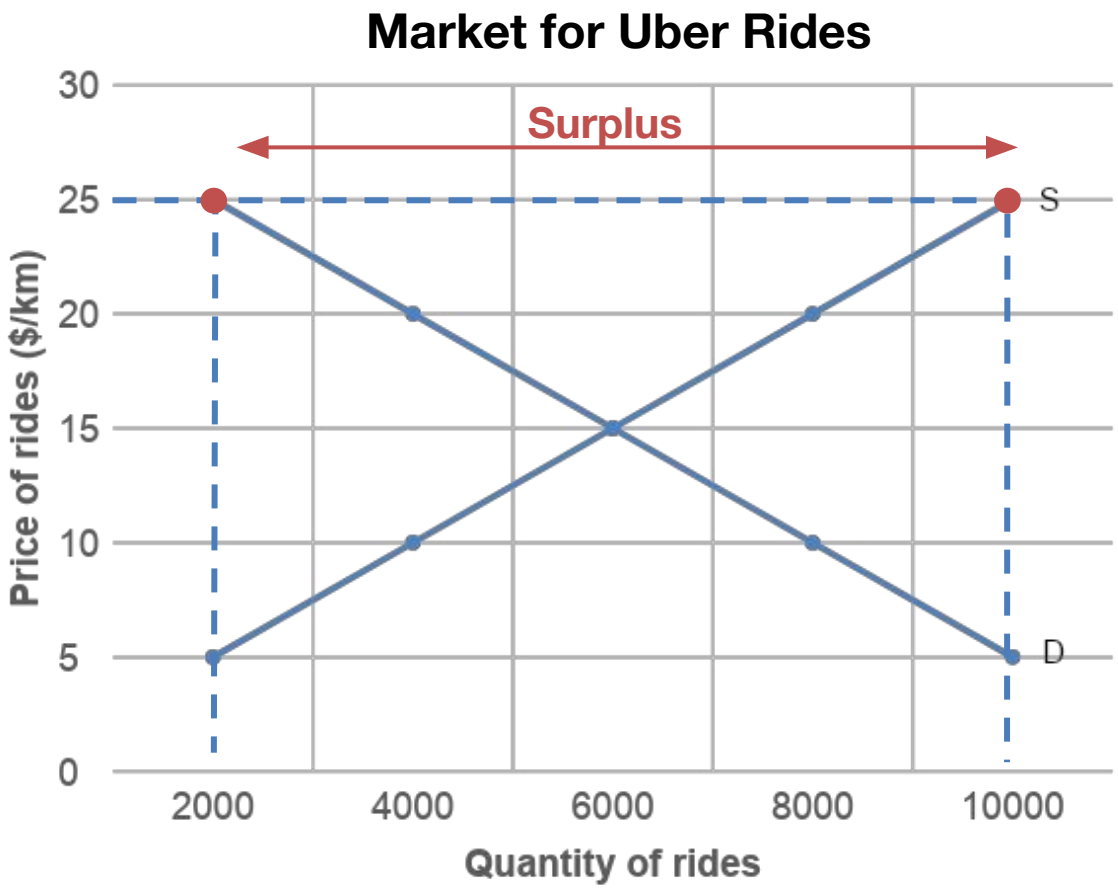


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Surplus – excess supply

Price of rides ($/km)	Quantity demanded (Riders)	Quantity supplied (Drivers)
25	2,000	10,000
20	4,000	8,000
15	6,000	6,000
10	8,000	4,000
5	10,000	2,000

If the price is higher than the equilibrium price, the quantity supplied is greater than quantity demanded, resulting in **excess supply (surplus)**.



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Key terms

Market equilibrium: occurs at a price where quantity demanded equals its quantity supplied.

The market has no shortages or surpluses.

Equilibrium price: Price at which quantity demanded equals quantity supplied.

Equilibrium quantity: Quantity at which quantity demanded equals quantity supplied.

Surplus (Excess supply): When quantity supplied exceeds quantity demanded.

Shortage (Excess demand): When quantity demanded exceeds quantity supplied.

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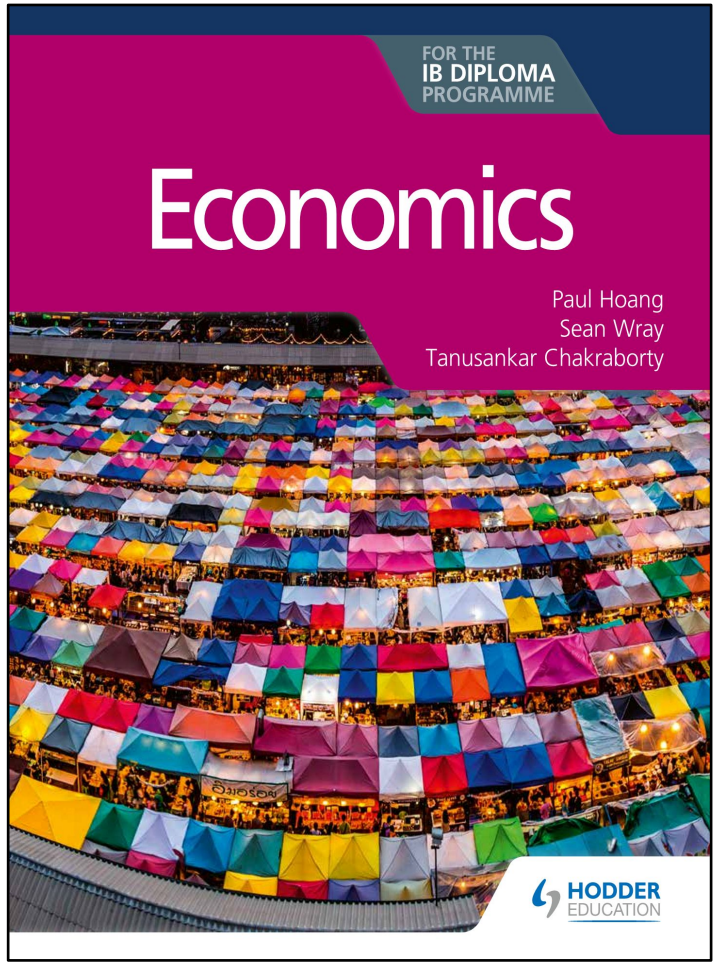
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Over to you…

Hoang, Wray, & Chakraborty (2020)

Economics for the IB Diploma Programme

- Page 70-71
- Paper 2 and 3 Exam Practice Question 5.1
- [4 marks]
- Paper 2 and 3 Exam Practice Question 5.2
- [2 marks]
- Paper 2 and 3 Exam Practice Question 5.3
- [5 marks]



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PAPER 2 AND 3 EXAM PRACTICE QUESTION 5.1

The diagram below represents the market for textbooks. Explain **two** possible reasons for the shift in the supply curve. *[4 marks]*

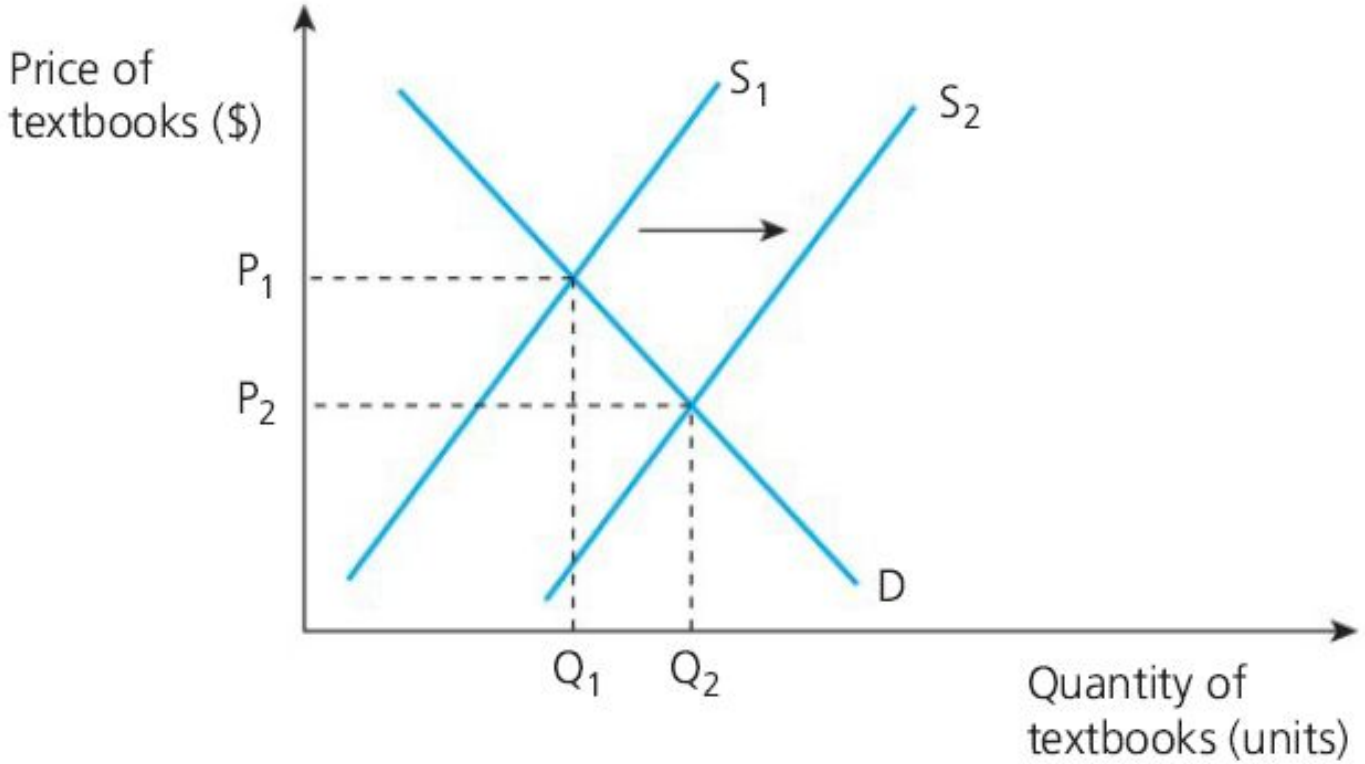


Figure 5.9 The market for textbooks

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[Paper 2 and 3 Exam Practice Question 5.1]

The supply curve has shifted to the right from S1 to S2, i.e. there is an increase in the supply of textbooks. Possible reasons for this include:

- Cheaper raw material costs, such as wood needed for the paper.
- Cheaper capital costs, such as printing equipment or printing machines.
- Cheaper transportation and distribution costs for publishers.
- A reduction in the per-unit tax imposed on textbooks.
- Technological progress, helping to reduce costs, such as e-textbooks and digital resources.
- Accept any other supply factor related to the textbook industry.

Mark as a *2* + *2*.

Award up to *2 marks* for explaining <u>each</u> cause of the rightward shift of the supply curve, up to the maximum of *4 marks*.

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PAPER 2 AND 3 EXAM PRACTICE QUESTION 5.2

With reference to the diagram below, calculate the change in total revenue if the market demand curve for grass seeds shifts from D_1 to D_2. *[2 marks]*

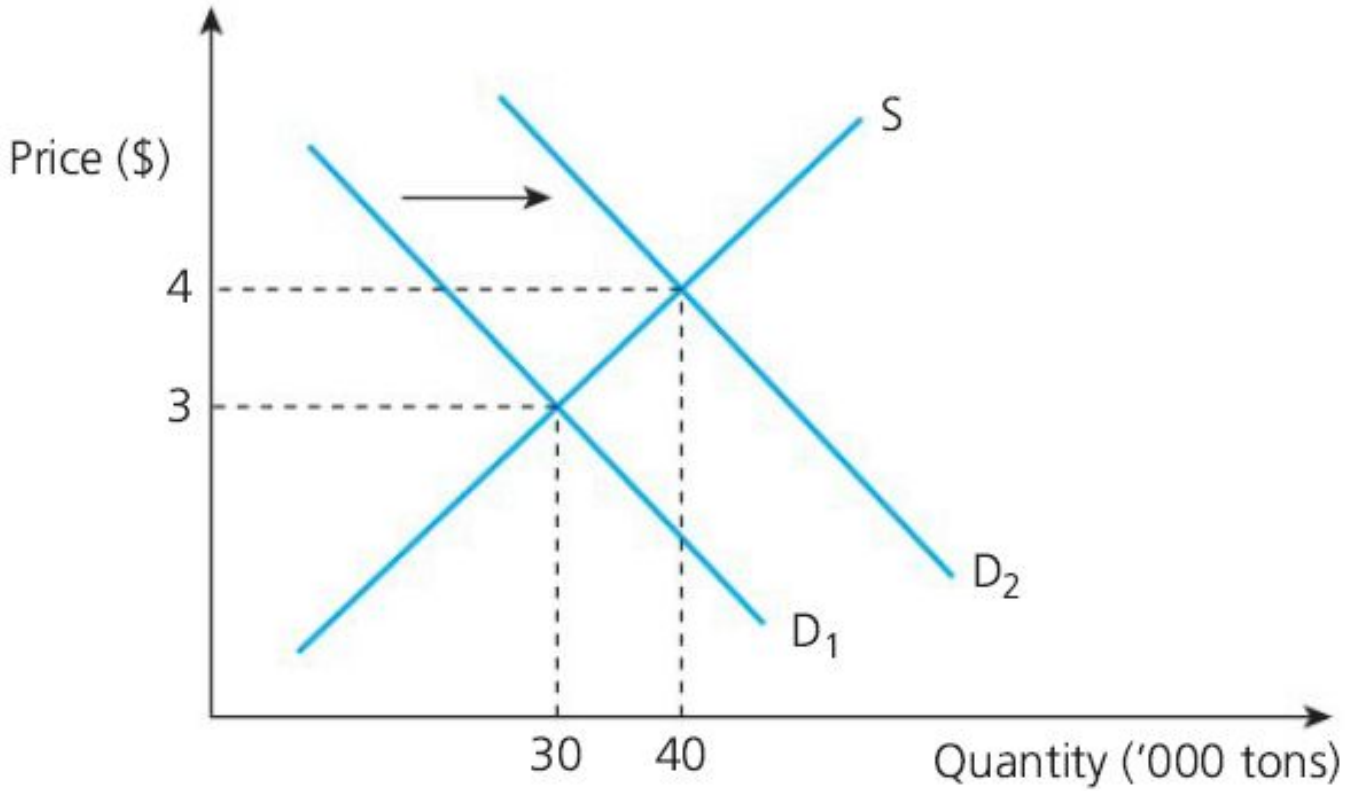


Figure 5.10 The market for grass seeds

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[Paper 2 and 3 Exam Practice Question 5.2]

 At P = $3 per ton, total revenue is $3 × 30,000 = $90,000

 At P = $4 per ton, total revenue increases to $4 × 40,000 = $160,000

 Therefore, the change in total revenue = **+$70,000**

Award *1 mark* for the correct answer and *1 mark* for showing accurate working out.

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PAPER 2 AND 3 EXAM PRACTICE QUESTION 5.3

The table below shows the demand and supply schedules for Product X.

P ($)	Q_d ('000)	Q_s ('000)
0	80	0
10	60	20
20	40	40
30	20	60
40	0	80

a Identify the equilibrium price. *[1 mark]*

b Calculate and identify the value of excess demand or supply if the price is $10. *[2 marks]*

c Calculate and identify the value of excess demand or supply if the price is $40. *[2 marks]*

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[Paper 2 and 3 Exam Practice Question 5.3]

a) $20 (as this is the price level where the quantity demanded equals the quantity supplied).

b) *Qd*($10) = 60,000 and *Qs*($10) = 20,000, so *ED*($10) = 40,000, i.e. when the price is $10, there is excess demand (a shortage) of 40,000 units.

c) *Qd*($40) = 0 and *Qs*($40) = 80,000, so *ES*($40) = 80,000, i.e. when the price is $40, there is excess supply (a surplus) of 80,000 units.

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PAPER 2 AND 3 EXAM PRACTICE QUESTION 5.4

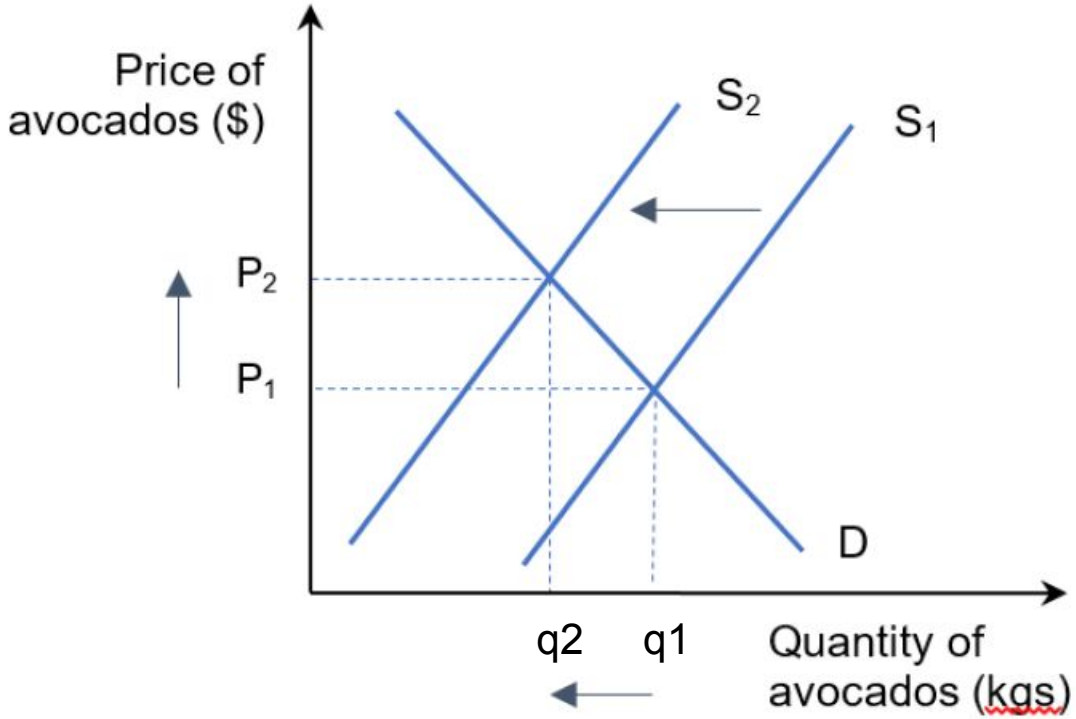
Mexico is the world's biggest producer of avocados, supplying about 45 per cent of the planet's avocados. In 2017, avocado prices reached a record high due to reduced harvests from major producers in Mexico, Peru and California. Prices also surged due to global demand in places like the US, Europe and China. The industry has become very profitable for Mexican farmers, with avocados being nicknamed Mexico's green gold.

Using a demand and supply diagram, explain why reduced harvests caused avocado prices to reach a record high in 2017. *[4 marks]*

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[Paper 2 and 3 Exam Practice Question 5.4]



Award up to *2 marks* for drawing a fully labelled diagram that shows a leftward shift of the supply curve of avocados.

Award up to another *2 marks* for an explanation, showing that a fall in the supply of avocados, due to the reduced harvests, causes the market price of avocados to rise.

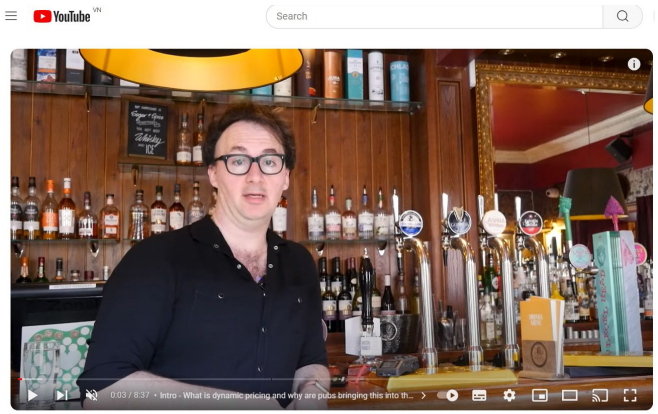
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Real World Example - Price Mechanism



Uber's surge pricing is an example of how the **price mechanism** reallocates resources.

The price mechanism uses **signals** and **incentives** to allocate resources via the forces of demand and supply in competitive markets.



Dynamic Pricing, Explained: AI business model | Uber Surge pricing

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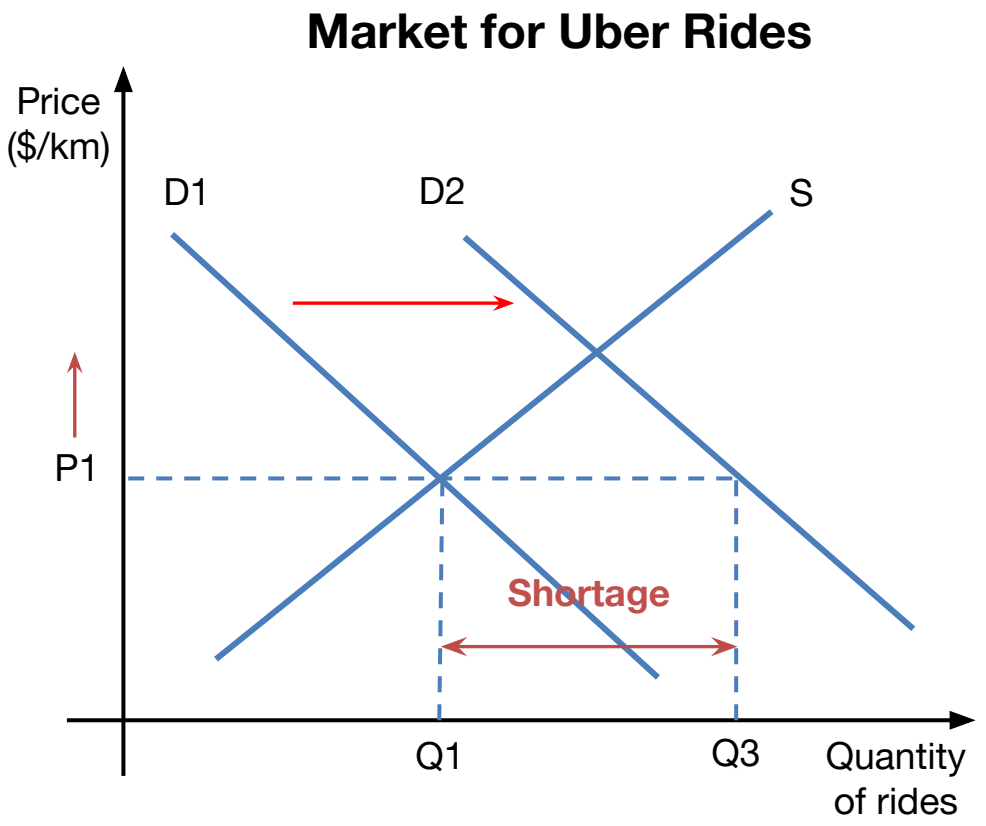
Real world example – price mechanism

Using the example of Uber, explain how the price mechanism allows a market to reach equilibrium following an increase in demand.

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Real World Example - Price mechanism

1. During peak periods, more people may demand Uber as a means of transport, increasing demand from D1 to D2.
2. Hence, at the regular Uber pricing of P1, excess demand (shortage) of Q3 - Q1 occurs.
3. Some consumers are willing and able to pay prices higher than P1 to secure an Uber ride. This exerts an upward pressure on price.
4. A higher price acts as a **signal** to producers (Uber drivers) and informs them that there are consumers (riders) who wish to get a ride at higher prices.

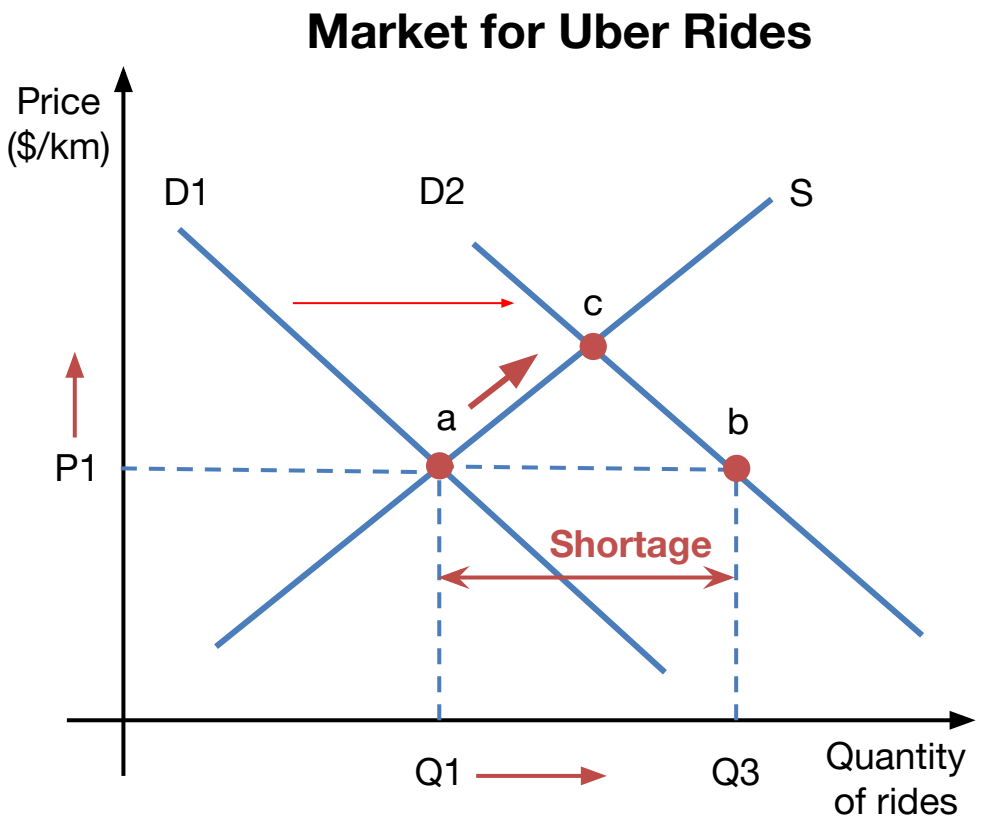


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Real World Example - Price mechanism

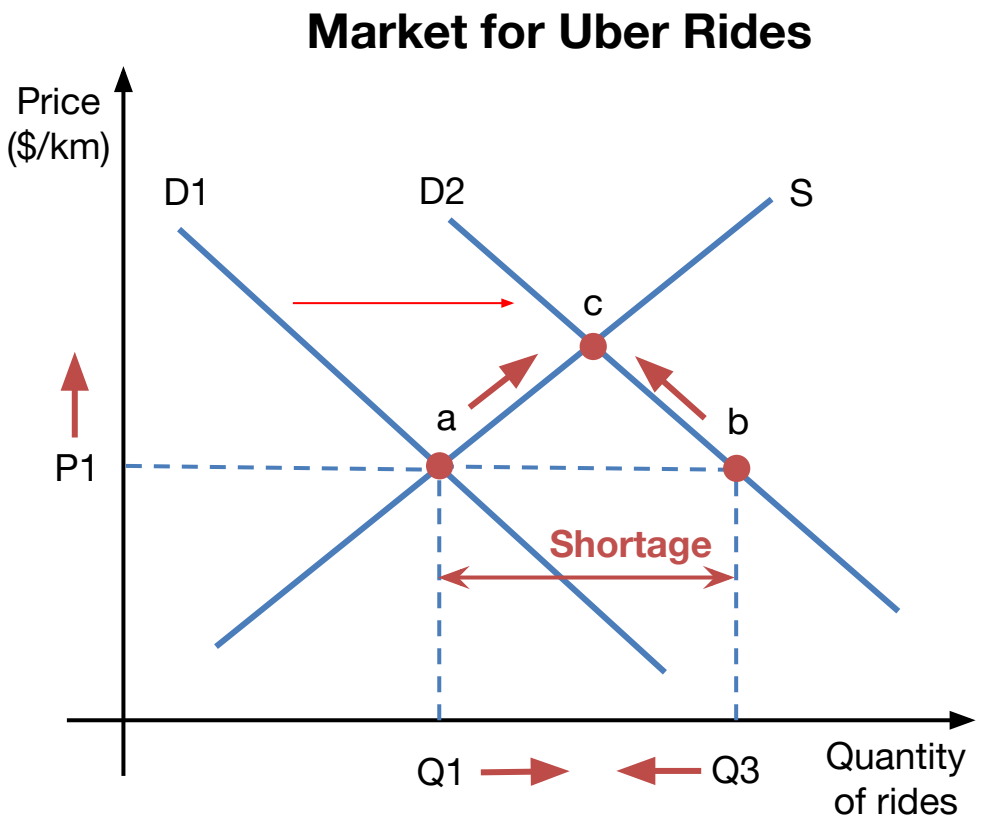
5. As price increases from P1, suppliers (Uber drivers) are **incentivized** to increase quantity supplied beyond Q1 due to the law of supply. More Uber drivers come to the area to provide their services.
6. Hence there is an upward movement along the supply curve from point *a* towards point *c.*



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Real World Example - Price mechanism

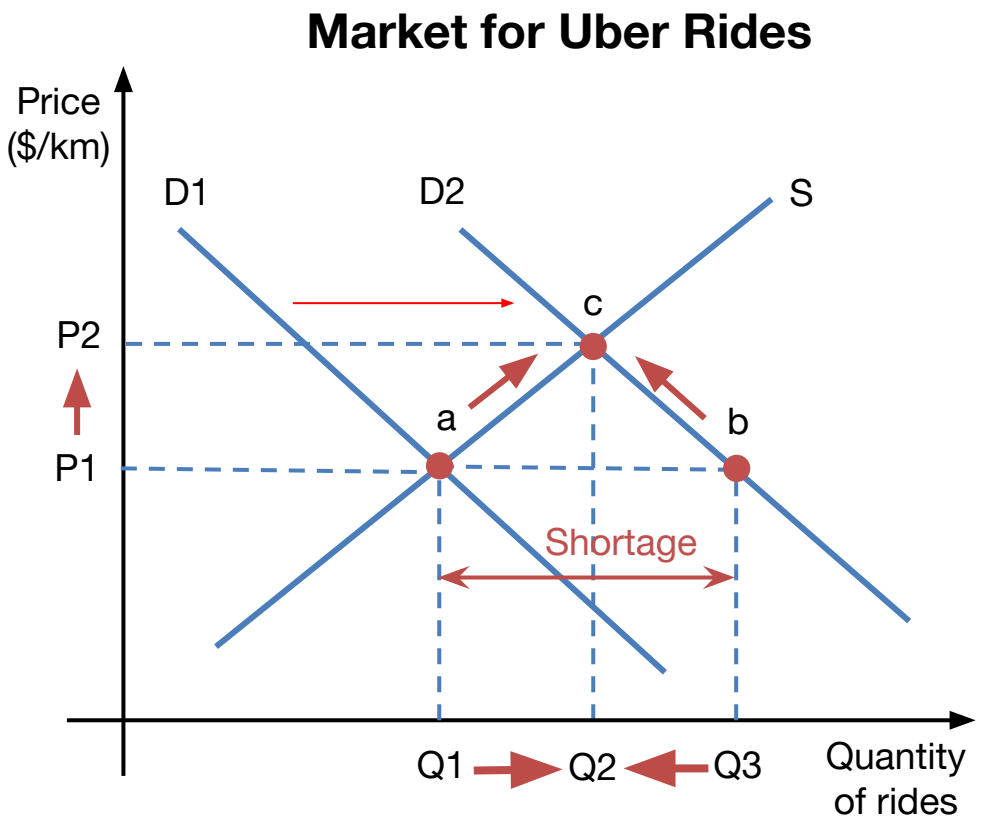
7. At the same time, as price increases from P1, some consumers (riders) are **disincentivized** to purchase rides, so quantity demanded falls from Q3 due to the law of demand. Some riders may opt for alternative forms of transport.
8. Hence there is an upward movement along the demand curve from point *b* towards point *c.*



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Real World Example - Price mechanism

9. The increase in price from P1 continues until quantity supplied equals to quantity demanded, at point *c*. A new market equilibrium forms at P2 and Q2. At this point, the market clears.
10. The price mechanism successfully **rations** resources to consumers who are willing to pay a higher price. Thus, it addresses the basic economic question of "to whom to produce for?"

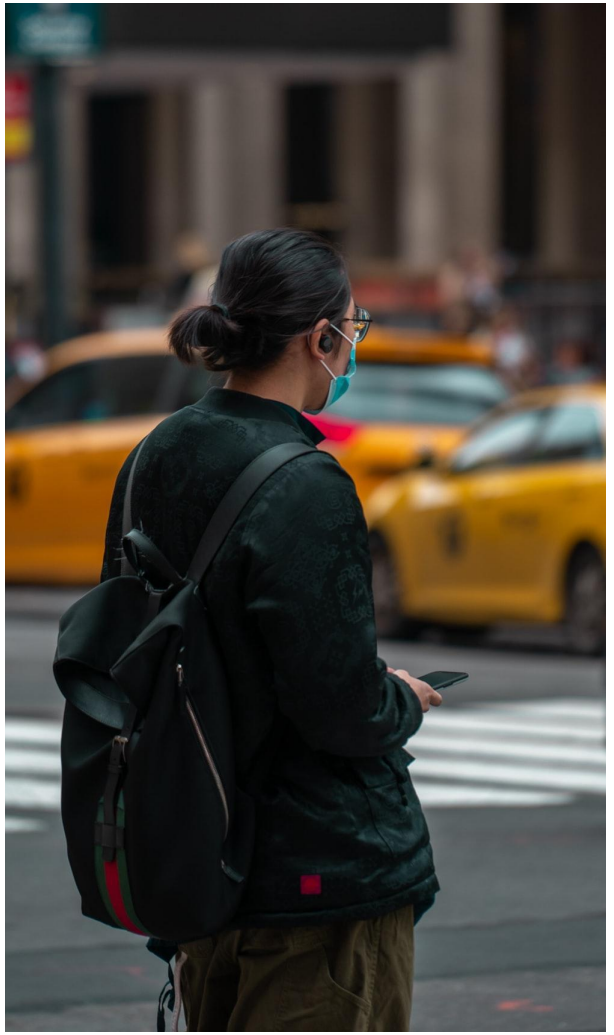


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Price mechanism – signals and incentives

Signaling function: price conveys market information to producers and consumers for making production and consumption decisions, in order to allocate resources.

Incentive function: price provides incentives for producers and consumers to change their behaviors to maximize their own benefits, in order to allocate resources.



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Price mechanism - rationing



The **rationing function** serves to **allocate scarce resources**, by increasing the market price to deter some consumers from buying the good.

An increase in price by the price mechanism helps to reduce quantity demanded, which is useful to eliminate shortages.

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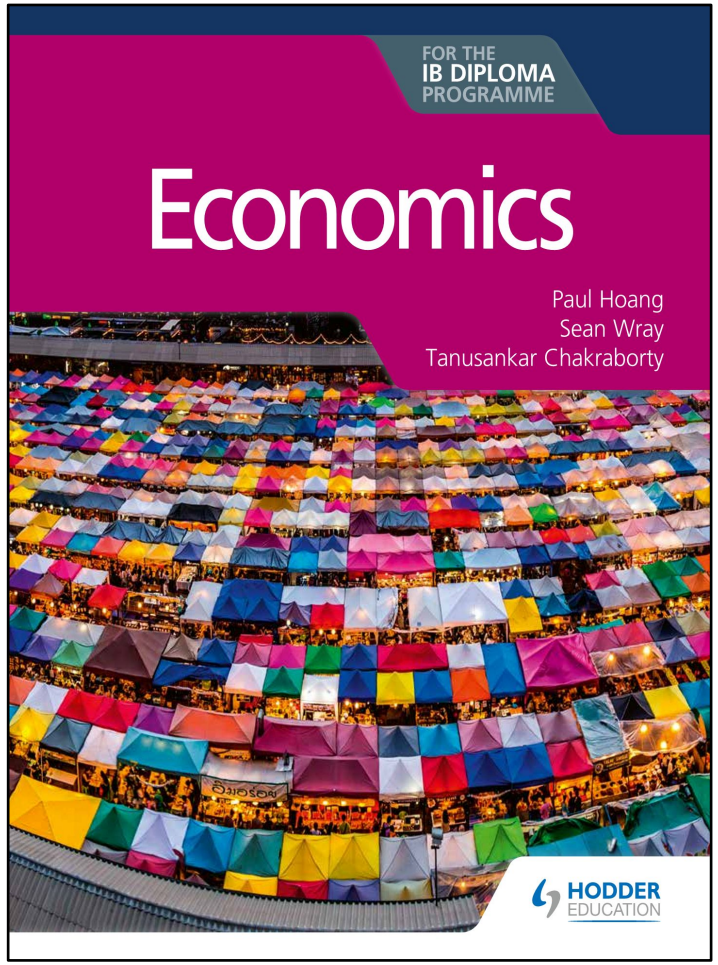
Over to you…

Hoang, Wray, & Chakraborty (2020)

Economics for the IB Diploma Programme

- Page 71
- Paper 2 and 3 Exam Practice Question 5.4
- [4 marks]

- Page 73
- Paper 1 Exam Practice Question 5.5
- [10 marks]



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PAPER 1 EXAM PRACTICE QUESTION 5.5

Using an appropriate diagram(s), explain the signalling and incentive functions of price. *[10 marks]*

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[Paper 1 Exam Practice Question 5.5]

Use the assessment criteria on page 1 of the level descriptors. Award up to the maximum marks as indicated. Answers may include:

 Definitions: signalling and incentive functions of the price mechanism.

 Explanations: the signalling function involves price adjustments to reflect relative scarcities and surpluses in product and resource markets for economic agents (producers and consumers), i.e. prices change to indicate where resources are required (price increases) and where they are not (price decreases).
The incentive function provides pricing information for economic agents, where a fall in price provides an incentive for consumers to purchase more, vice versa.
Explanation that for competitive markets to work efficiently, economic agents must respond to price signals and incentives in product and resource markets.

 Diagram: Demand and supply diagram(s) showing the impact of the signalling and incentive functions of price and quantity for particular markets, illustrating the impact of the signalling and incentive functions of price.

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Real World Example - Essay

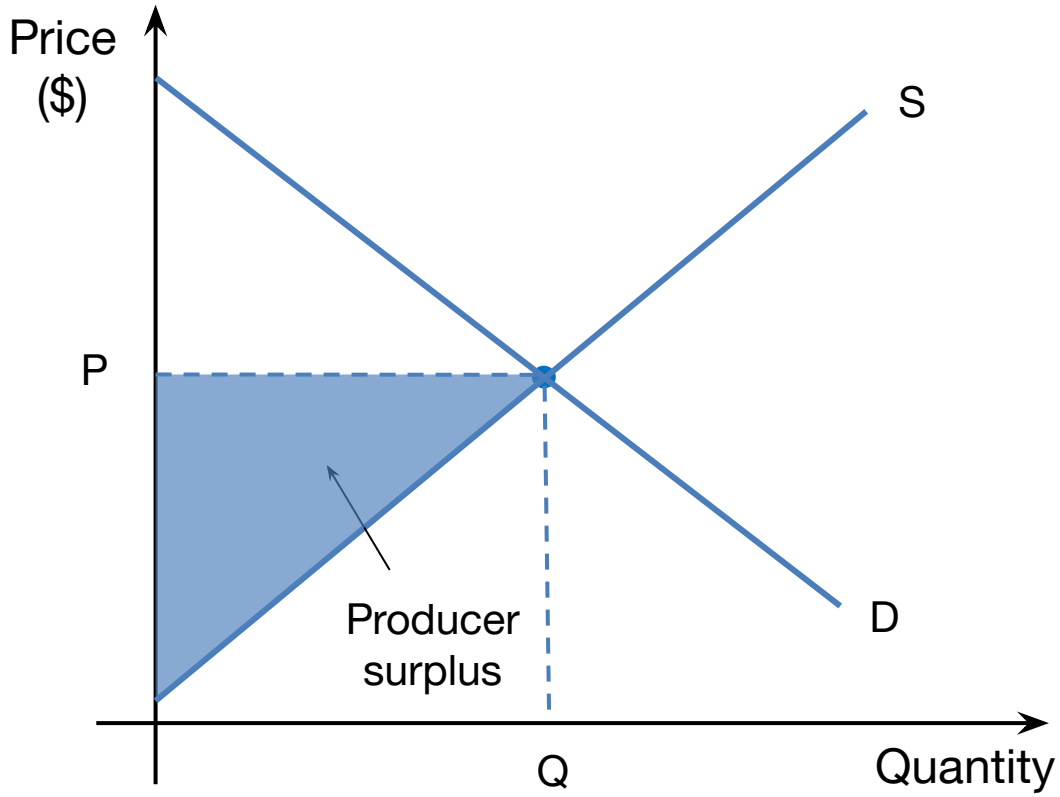
Video: 2 New Low-Cost Airlines Launch as Americans Return To The Skies

Using the airline industry as an example, explain how the price mechanism works to reallocate resources in a market following an increase in market supply.

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Producer surplus

Producer surplus is the positive difference between the price that producers receive from selling a good and the minimum amount they are prepared to sell the good at.



Producer surplus is identified by the area below the selling price and above the supply curve for the quantities sold.

(HL only)

Producer surplus can be calculated by the area of the triangle below P and above the supply curve.

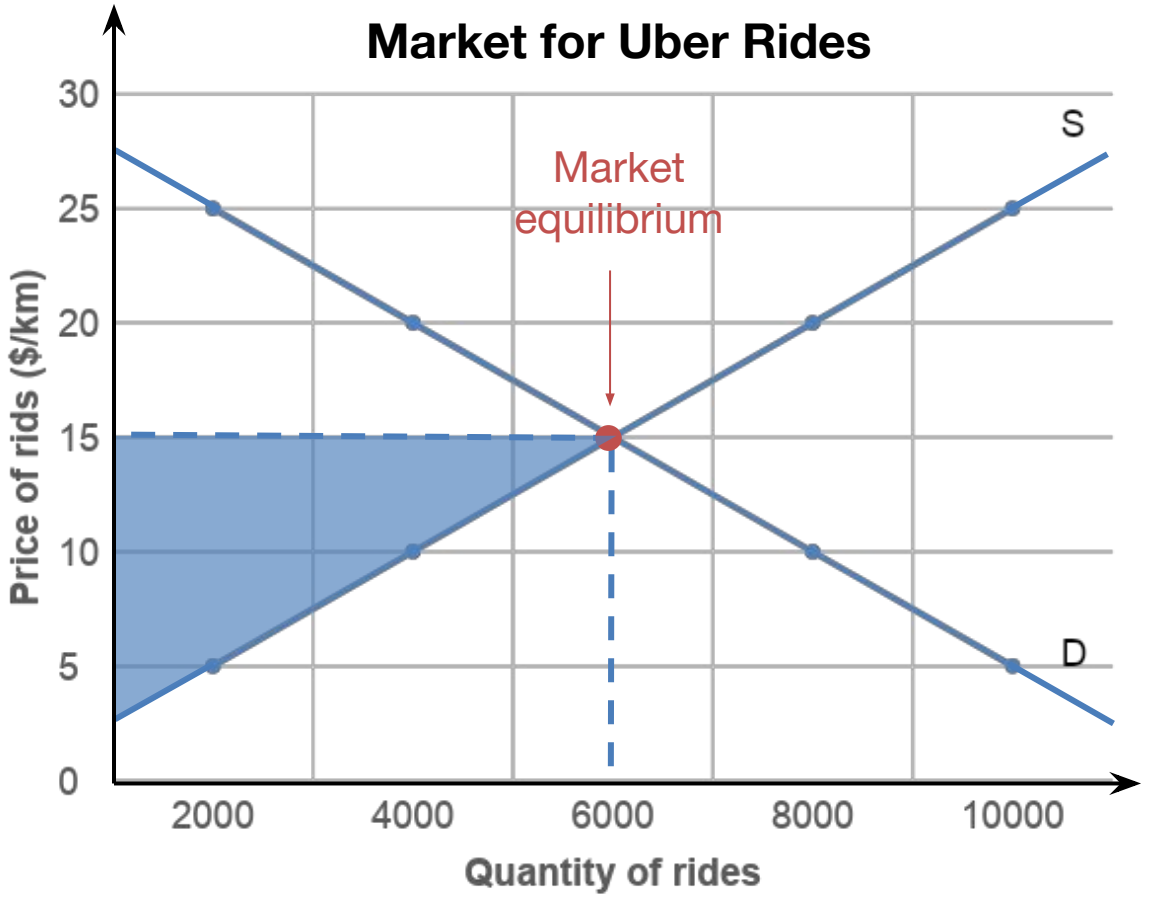
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Producer surplus

Calculate the producer surplus in the Uber market.

$$\text{Producer surplus} = \frac{(\$15 - \$2.5) \times 6{,}000}{2}$$

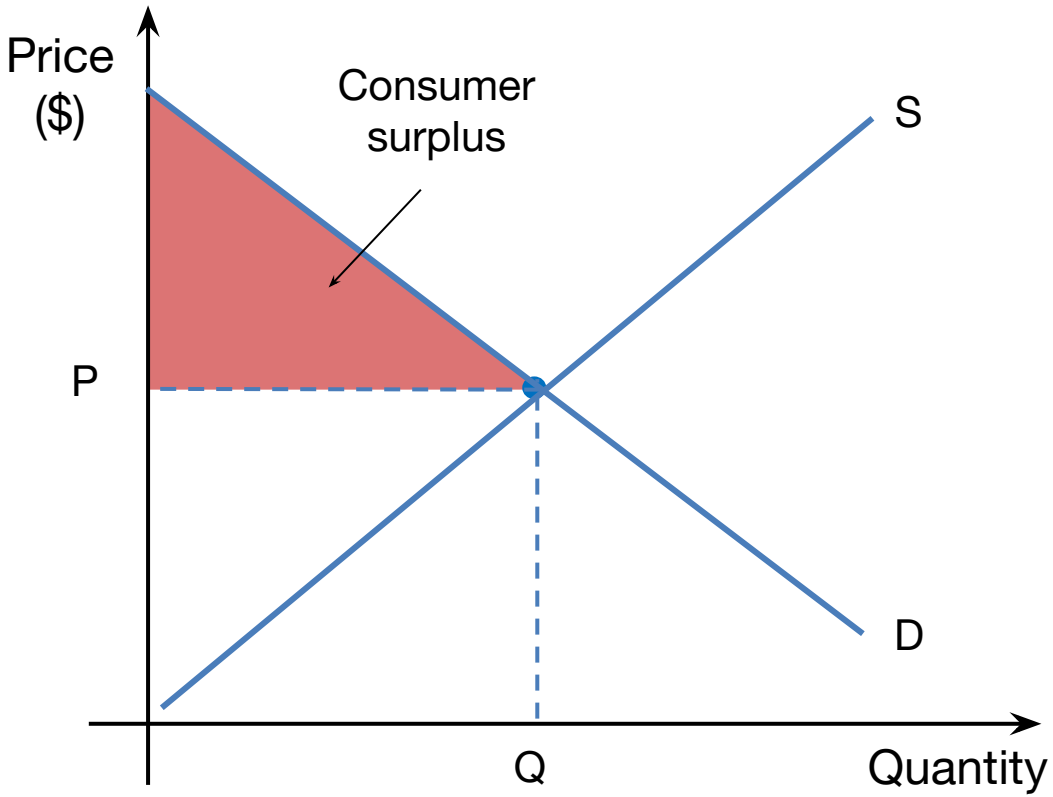
Producer surplus = $37,500



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Consumer surplus

Consumer surplus is the positive difference between the amount that a consumer is willing and able to pay for a good and the amount they actually pay.



Consumer surplus is identified by the area above the buying price & below the demand curve for the quantities purchased.

(HL only)

Consumer surplus can be calculated by the area of the triangle above P and below the demand curve.

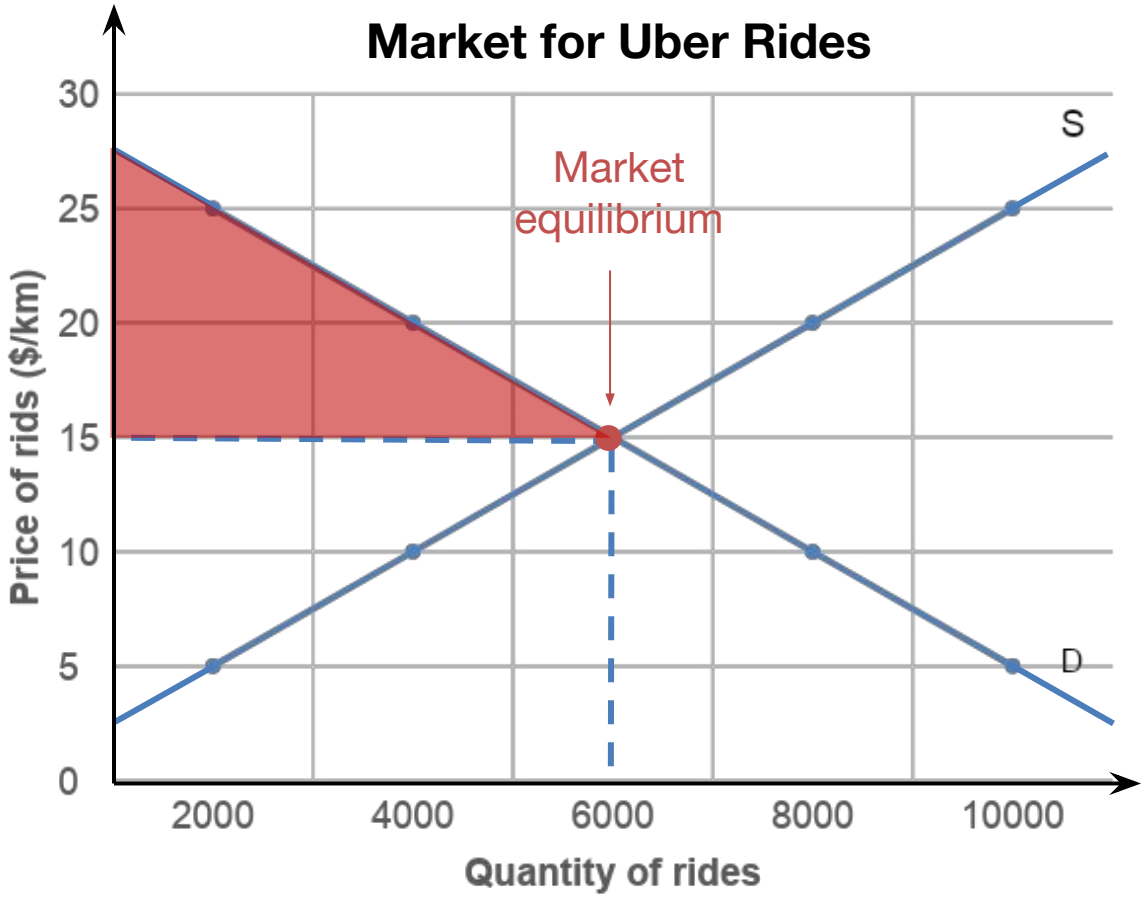
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Consumer surplus

Calculate the consumer surplus in the Uber market.

$$\text{Consumer surplus} = \frac{(\$27.5 - \$15) \times 6{,}000}{2}$$

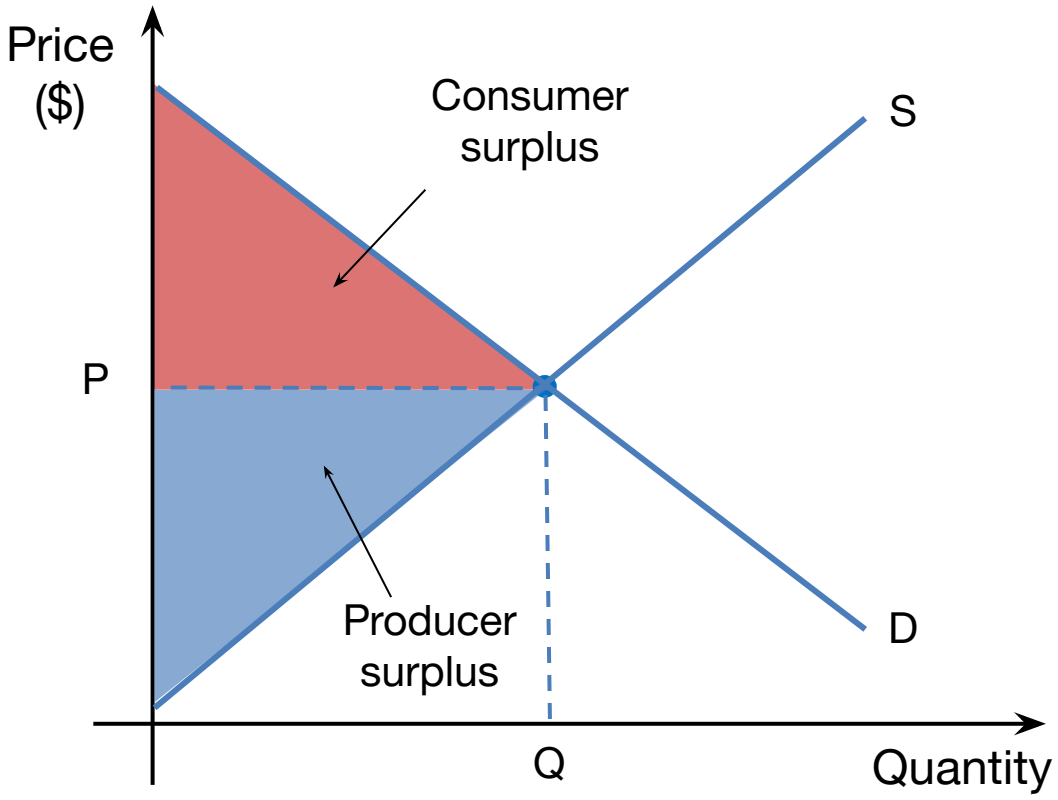
Consumer surplus = $37,500



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Social surplus (Community surplus) and allocative efficiency

Social surplus is the sum of consumer surplus and producer surplus at a particular price and quantity.

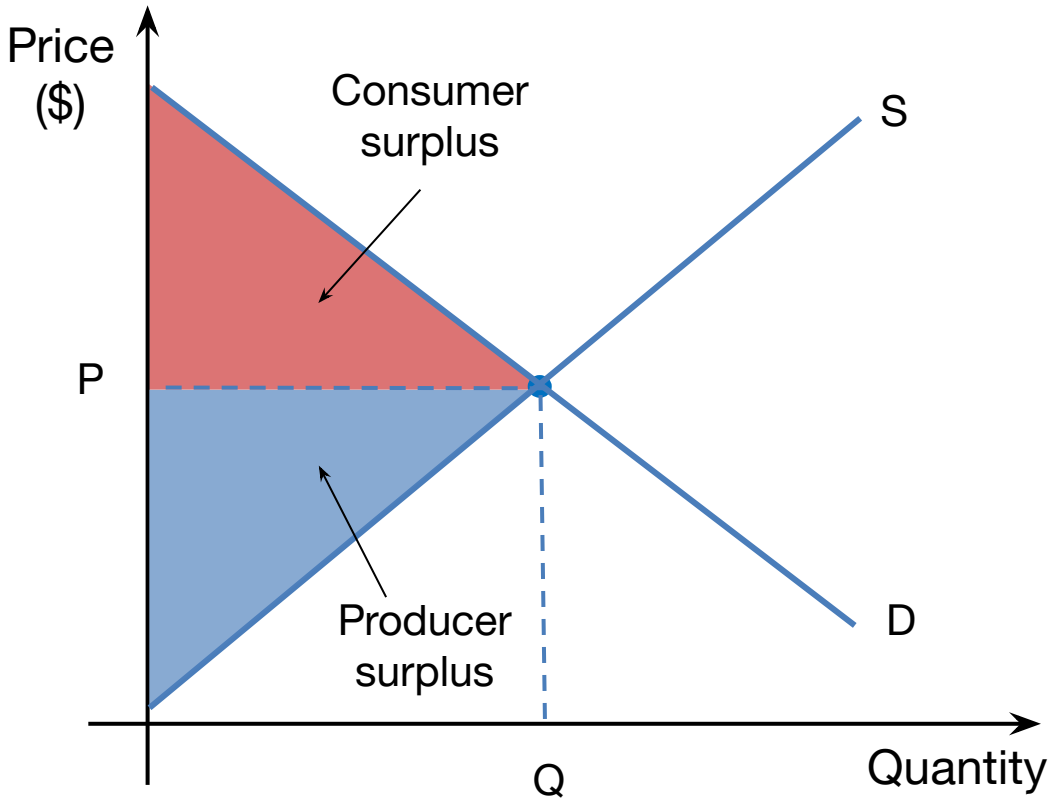


Social surplus is maximized at the market equilibrium.

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Social surplus (Community surplus) and allocative efficiency

Allocative efficiency is the socially optimum outcome where resources are allocated such that the sum of consumer and producer surplus are maximized.

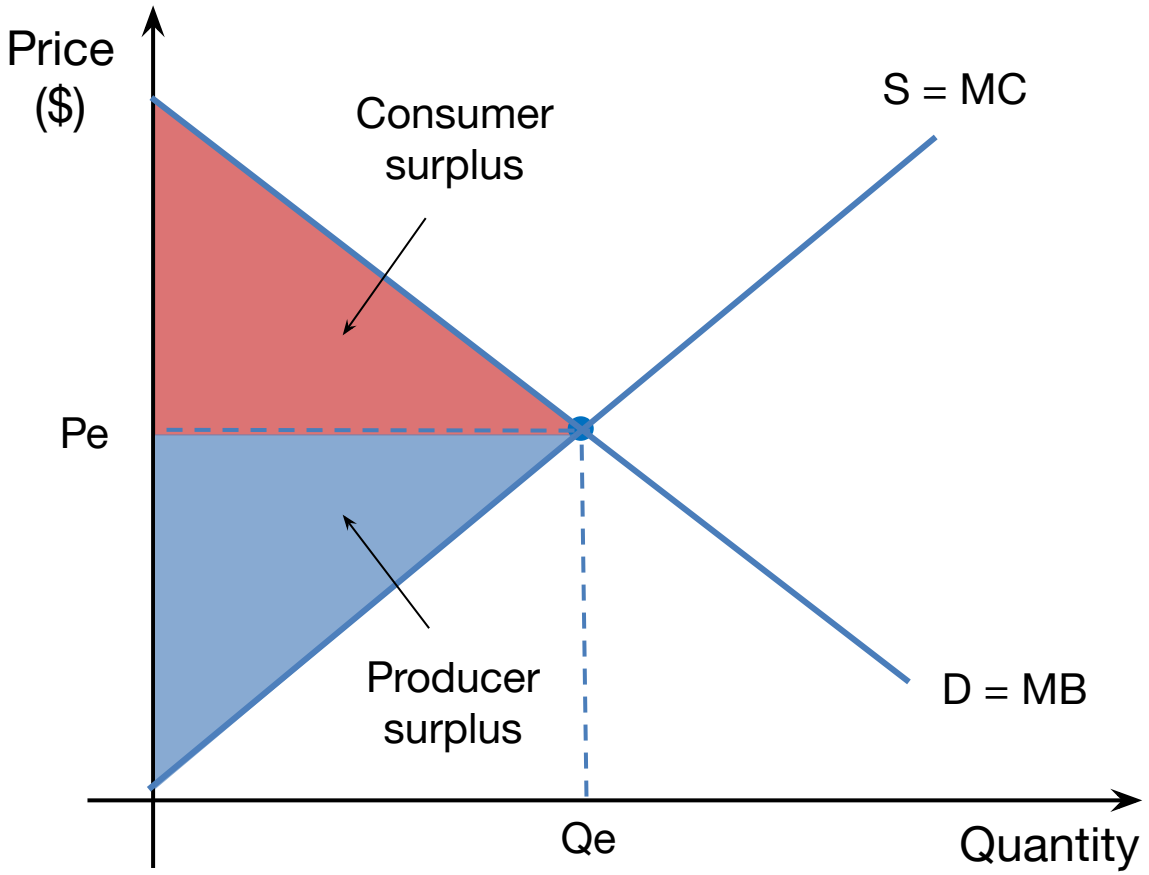


In other words, no one can be better-off without making someone else worse-off. This is attained at the competitive market equilibrium.

How would social surplus be shown if producers charged a price above the equilibrium?

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Marginal benefits and marginal costs



The demand curve can also be considered the **marginal benefit curve**.

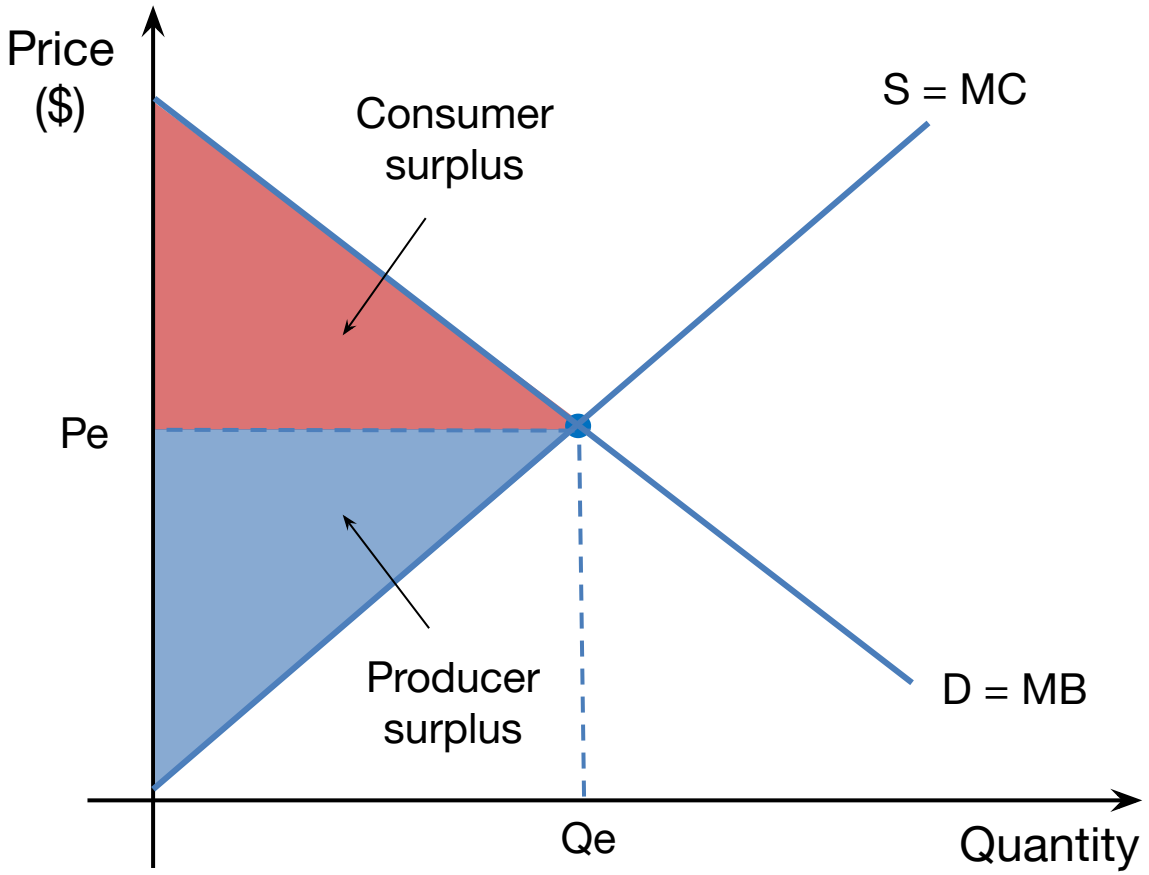
Why does marginal benefit decrease as quantity increase?

The supply curve can also be considered the **marginal cost curve**.

Why does marginal cost increase as quantity increase?

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Marginal benefits and marginal costs



Marginal cost: The additional cost incurred by producers when they produce an additional unit.

Marginal benefit: The additional benefit enjoyed by consumers when they consume an additional unit.

Allocative efficiency is attained when MC = MB

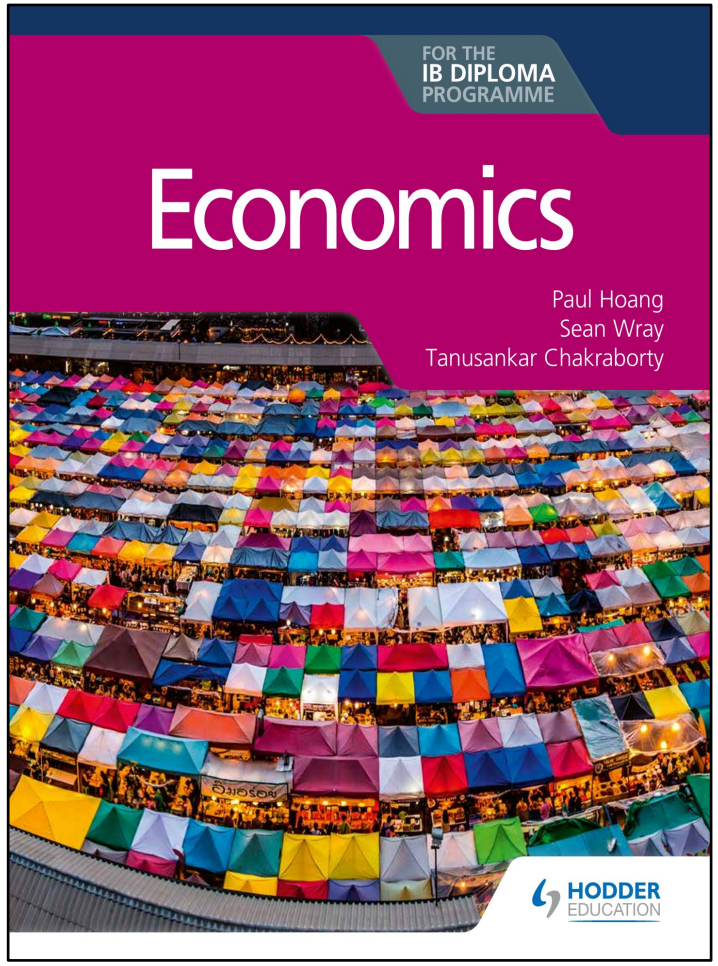
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Hoang, Wray, & Chakraborty (2020)

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- Page 76
- Paper 3 Exam Practice Question 5.6
- [12 marks]
- Paper 3 Exam Practice Question 5.7
- [10 marks]



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PAPER 3 EXAM PRACTICE QUESTION 5.6 (HL ONLY)

a Use the data in the demand and supply schedules below to plot a demand and supply diagram. *[4 marks]*

P ($)	Qd	Qs
0	80	0
10	60	20
20	40	40
30	20	60
40	0	80

b Use your graph to calculate the total value of the consumer surplus at the equilibrium price. *[2 marks]*

c Calculate the total value of the producer surplus if the price falls to $10. *[2 marks]*

d Calculate the quantity of excess demand at a price of $10 per unit. *[2 marks]*

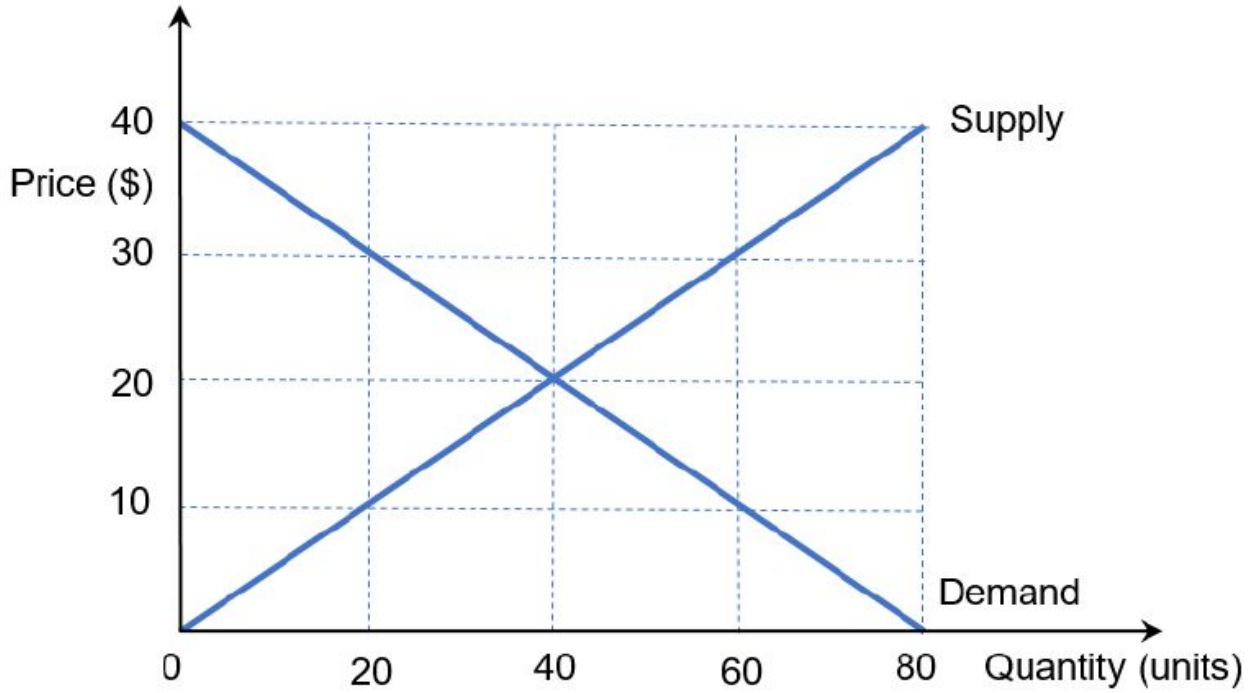
e Calculate the quantity of excess supply at a price of $30 per unit. *[2 marks]*

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[Paper 3 Exam Practice Question 5.6]

a)



Award *1 mark* each for the following: *y*-axis label, *x*-axis label, supply curve, and demand curve (which must be plotting accurately and labelled accordingly), up to the maximum of *4 marks*.

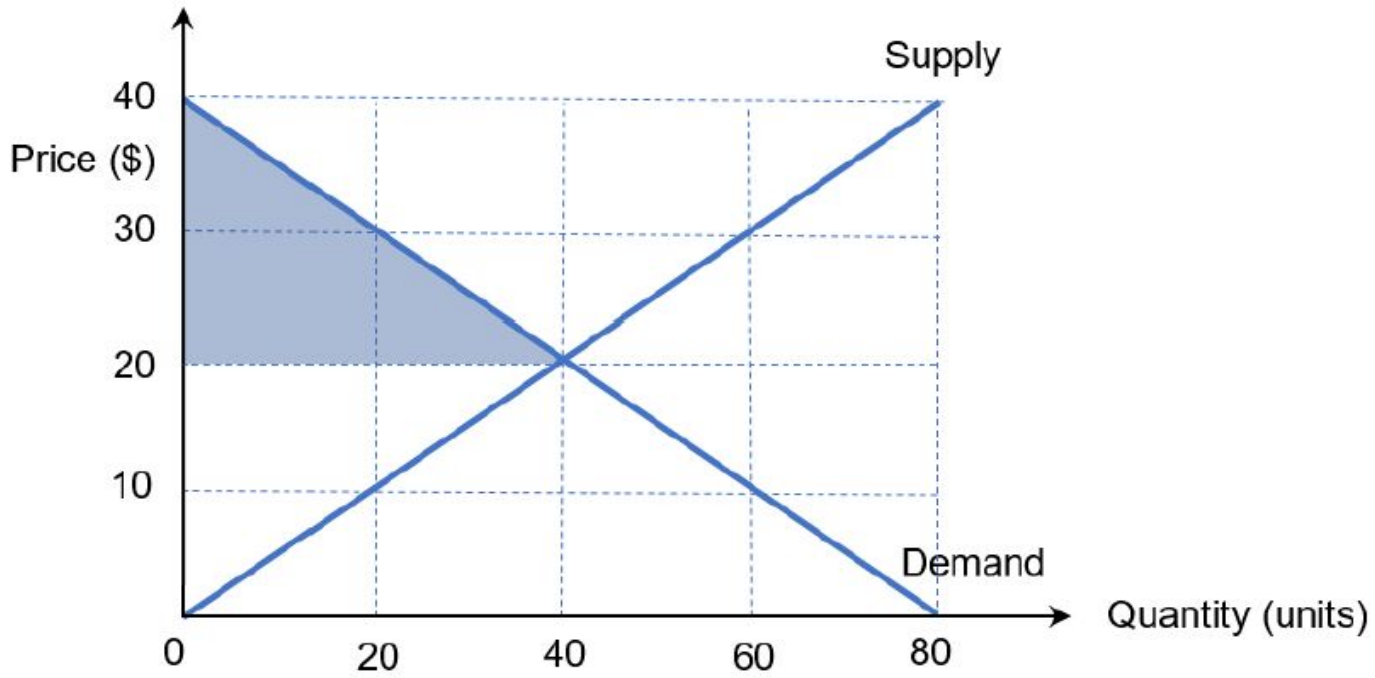
There is no need to include a title for this diagram.

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b) Consumer surplus exists when there are customers willing and able to pay more than the equilibrium price. Hence at $20, the consumer surplus is shown by the triangular area under the demand curve, above the equilibrium price level.

Hence, the value of consumer surplus is [($40 – $20) × 40]/2 = **$400**

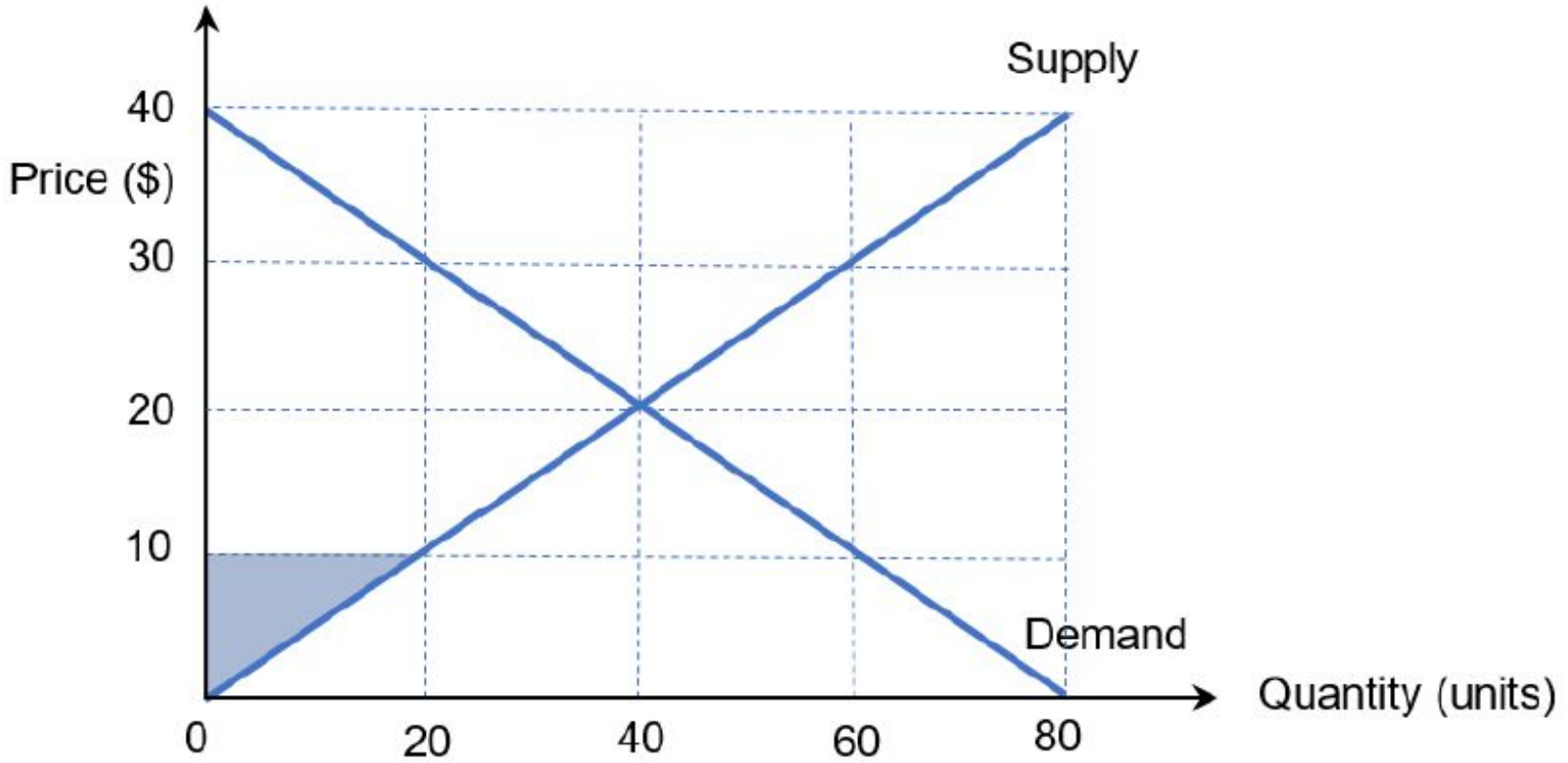


Award *1 mark* for the correct answer and *1 mark* for showing the working out.

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c) The value of producer surplus at $10 is [($10 – $0) × 20]/2 = **$100**



Award *1 mark* for the correct answer and *1 mark* for showing the working out.

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d) From the graph in Question b, it can be seen that:

- If price is $10, demand = 60 and supply = 20
- Hence, there is excess demand of 60 – 20 = **40 units**.

Award *1 mark* for the correct answer and *1 mark* for showing the working out.

e) From the graph in Question b, it can be seen that:

- At a price of $30, demand = 20 units and supply = 60 units
- Thus, there is excess supply of **40 units**.

Award *1 mark* for the correct answer and *1 mark* for showing the working out.

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PAPER 3 EXAM PRACTICE QUESTION 5.7 (HL ONLY)

Explain the view that the best allocation of resources occurs when consumer surplus and producer surplus are maximized. *[10 marks]*

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[Paper 1 Exam Practice Question 5.7]

Use the assessment criteria on page 1 of the level descriptors. Award up to the maximum marks as indicated. Answers may include:

- Definitions: consumer surplus and producer surplus.
- Explanations: of how consumer and producer surplus are maximized at the free market equilibrium price, and how disequilibrium brings inefficiencies and opportunity costs. For example, at a price lower than the equilibrium, consumer surplus will increase but at the cost of lower producer surplus, and vice versa for an increase in price above the equilibrium. Explanation of the economic interpretation of 'best allocation of resources', i.e. economic efficiency.
- Diagrams: changes in consumer and producer surplus following changes in prices, perhaps caused by factors such as changes in indirect taxes and producer subsidies from the domestic government.

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Assessment criteria

Paper 1

Section A = 10 marks

Marks	Level descriptor
0	• The work does not reach a standard described by the descriptors below.
1–2	• The response indicates little understanding of the specific demands of the question. • Economic theory is stated but it is not relevant. • Economic terms are stated but they are not relevant.
3–4	• The response indicates some understanding of the specific demands of the question. • Relevant economic theory is described. • Some relevant economic terms are included.
5–6	• The response indicates understanding of the specific demands of the question, but these demands are only partially addressed. • Relevant economic theory is partly explained. • Some relevant economic terms are used appropriately. • Where appropriate, relevant diagram(s) are included.
7–8	• The specific demands of the question are understood and addressed. • Relevant economic theory is explained. • Relevant economic terms are used mostly appropriately. • Where appropriate, relevant diagram(s) are included and explained.
9–10	• The specific demands of the question are understood and addressed. • Relevant economic theory is fully explained. • Relevant economic terms are used appropriately throughout the response. • Where appropriate, relevant diagram(s) are included and fully explained.

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Test your knowledge on this unit: Kahoot!

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